



React Hooks for SPFx: Build Cleaner, Faster, Better

Don Kirkham - Microsoft MVP, M365 Development & Microsoft Graph



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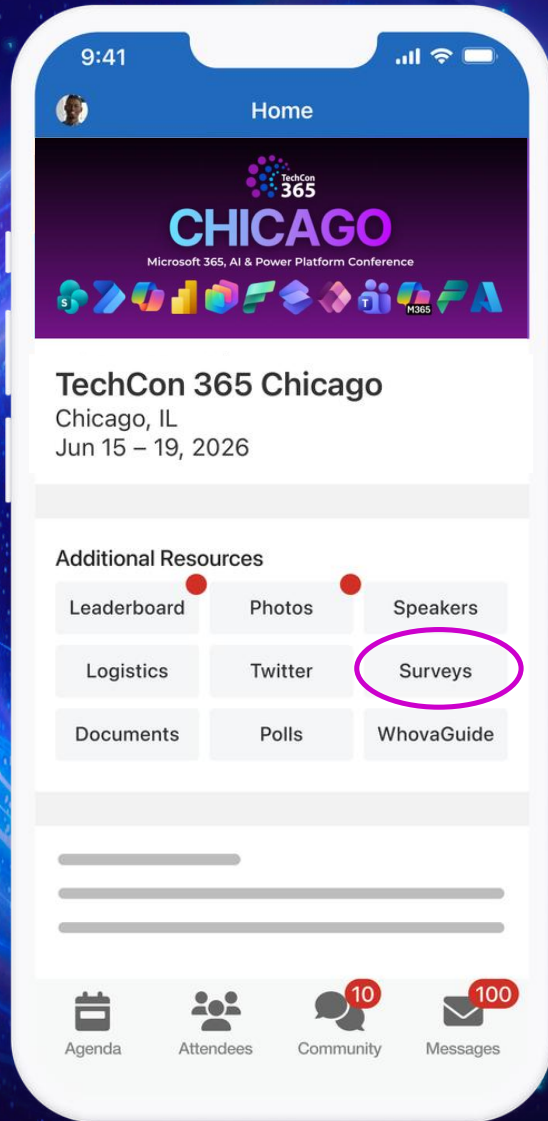
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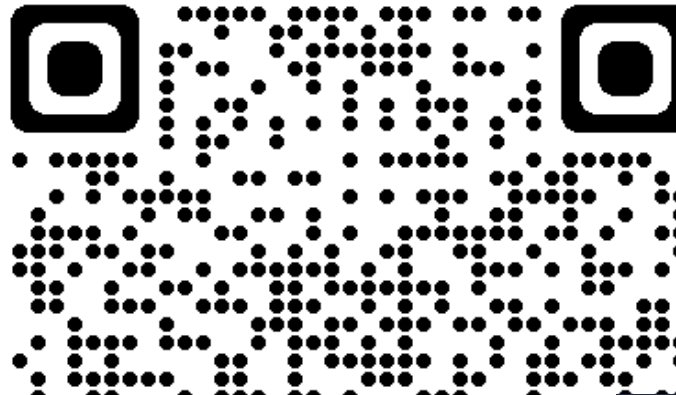
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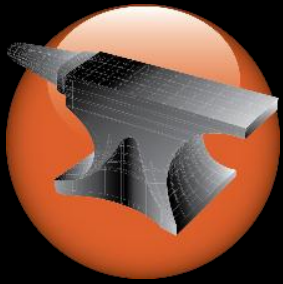


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- Enterprise portals, dashboards, and reporting for operational visibility
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PICTURE TIME!

Thursday June 26, 2025
1:33 PM
Room 344

Today's Activities

- 12:40 PM - 01:50 PM
Power BI Storage Models: The Ultimate Showdown - Philip Seamark
- 02:00 PM - 03:10 PM
Enhancing SPFx projects with Live Data integration to SharePoint, MS Graph and beyond - Don Kirkham
- 04:00 PM - 05:10 PM
SSMS 21 Spotlight: What's new and why it matters - Makena Barickman

04:00 PM - 05:10 PM
Enhancing SPFx projects with Live Data integration to SharePoint, MS Graph and beyond - Don Kirkham

04:00 PM - 05:10 PM
SSMS 21 Spotlight: What's new and why it matters - Makena Barickman

AGENDA

- What are Function(al) Components and Hooks
- Classes vs Function Components
- Converting your SPFx project
- Hooks, Hooks, Hooks!

FUNCTIONAL COMPONENTS + HOOKS (FC+HOOKS)

THE HISTORY

- 2015 – FCs introduced as Class alternative
 - Stateless
- 2019 – React Hooks introduced in React 16.8
 - Allowed granular state management in FC
- SharePoint Framework (SPFx)
 - Microsoft Yeoman Generator
 - Default scaffolds with Class Components
 - FC+Hooks only available in SPFx 1.9+
- Function vs Functional
 - Functional confusing based on Functional Computing
 - Functional Component is a defined React element
 - Function results in cleaner code
 - Functional uses implicit properties (inconsistent)
 - “Preferred” way to do function-based components

CLASS COMPONENT (DEFAULT SPFX)

```
1 > import * as React from 'react'; ...
3
4 > export interface ILifecycleClassicProps { ...
10 }
11
12 > export interface ILifecycleClassicState { ...
16 }
17
18 > export default class LifecycleClassic extends React.Component<ILifecycleClassicProps, ILifecycleClassicState> {
19
20   protected timer?: number;
21
22   // Constructor: initialize state and bind handlers
23   > constructor(props: ILifecycleClassicProps) { ...
27   }
28
29   // componentDidMount: log mount event
30   > componentDidMount(): void { ...
32   }
33
34   // componentDidUpdate: log updates and handle count logic
35   > componentDidUpdate(prevProps: ILifecycleClassicProps, prevState: ILifecycleClassicState): void { ...
47   }
48
49   // componentWillUnmount: log unmount and clean up
50   > componentWillUnmount(): void { ...
53   }
54
55   // #region other functions ...
99
100 > render(): JSX.Element { ...
122   }
123 }
```

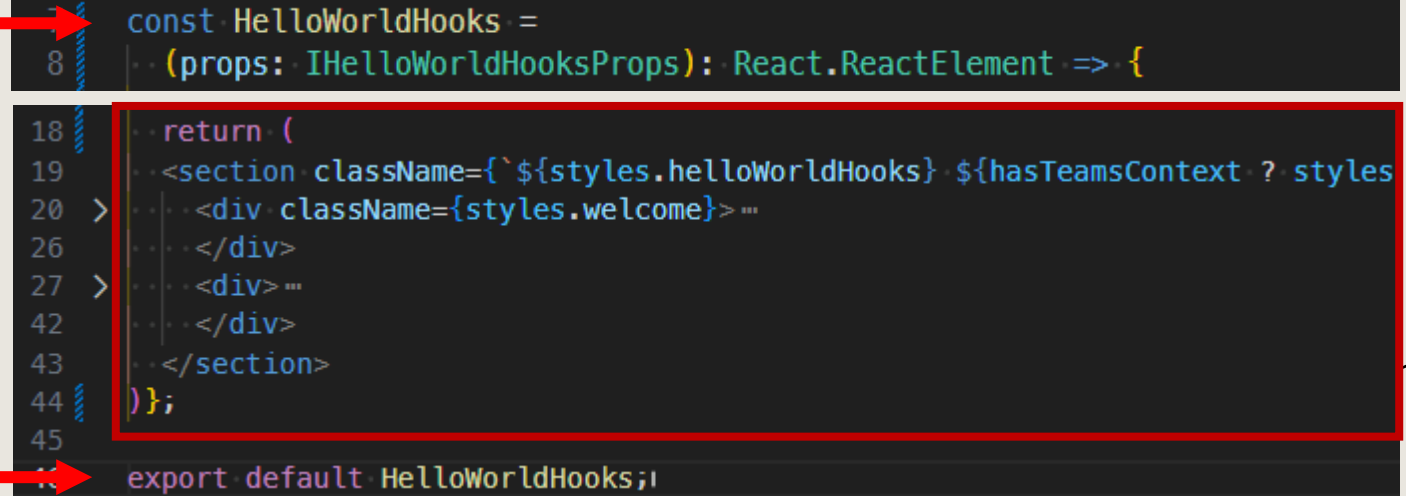
The diagram illustrates the structure of a Class Component in SPFX. It shows the following components:

- Interfaces:** `ILifecycleClassicProps` and `ILifecycleClassicState` are defined at the top.
- Class Declaration:** `LifecycleClassic` is a class that extends `React.Component<ILifecycleClassicProps, ILifecycleClassicState>`.
- Lifecycle Methods:** The class implements several lifecycle methods: `constructor`, `componentDidMount`, `componentDidUpdate`, and `componentWillUnmount`.
- Render Method:** The `render` method is defined at the bottom.

Red boxes and arrows highlight the key parts of the code: the interfaces, the class declaration, the lifecycle methods, and the render method.

WHAT ARE FUNCTION COMPONENTS?

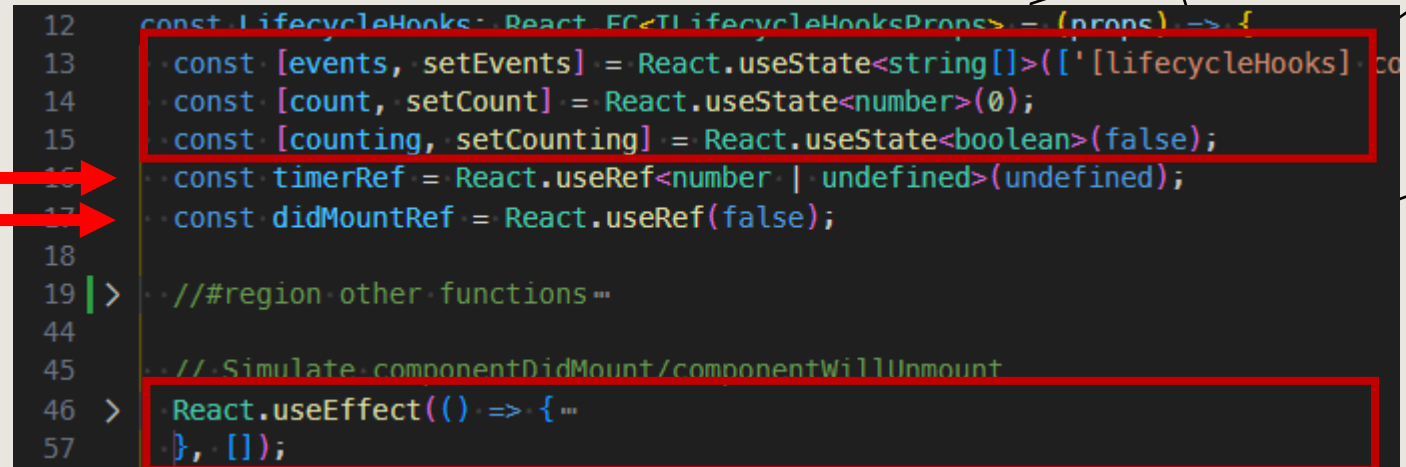
- Simply, JavaScript Functions
 - Accepts Properties
 - Stateless
- Output: React Element (JSX/TSX)
 - Just like render function in class



```
7 const HelloWorldHooks =  
8   (props: IHelloWorldHooksProps): React.ReactElement => {  
18     return (  
19       <section className={` ${styles.helloWorldHooks} ${hasTeamsContext ? styles.  
20     > <div className={styles.welcome}> ...  
26     </div>  
27     <div> ...  
42     </div>  
43     </section>  
44   );  
45  
46   export default HelloWorldHooks;
```

WHAT ARE HOOKS?

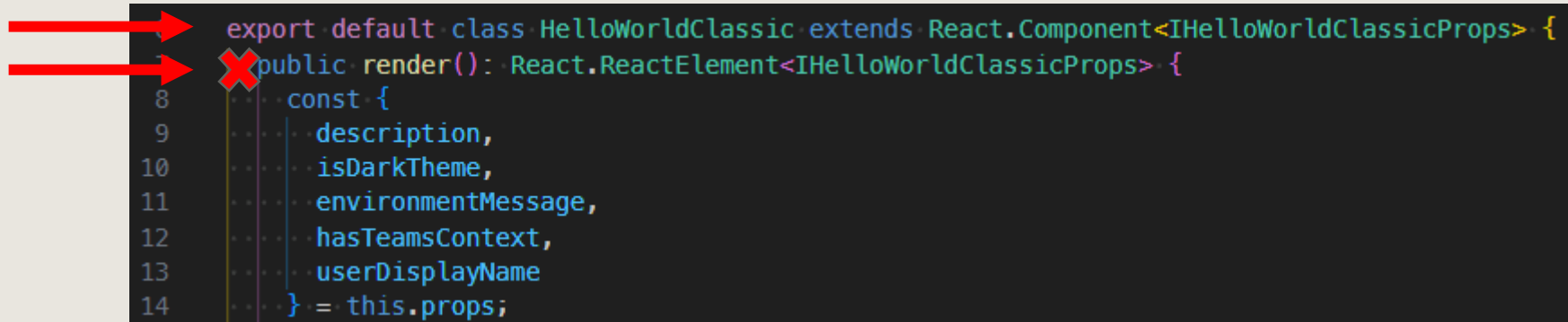
- Primarily used to manage state
- So much more!



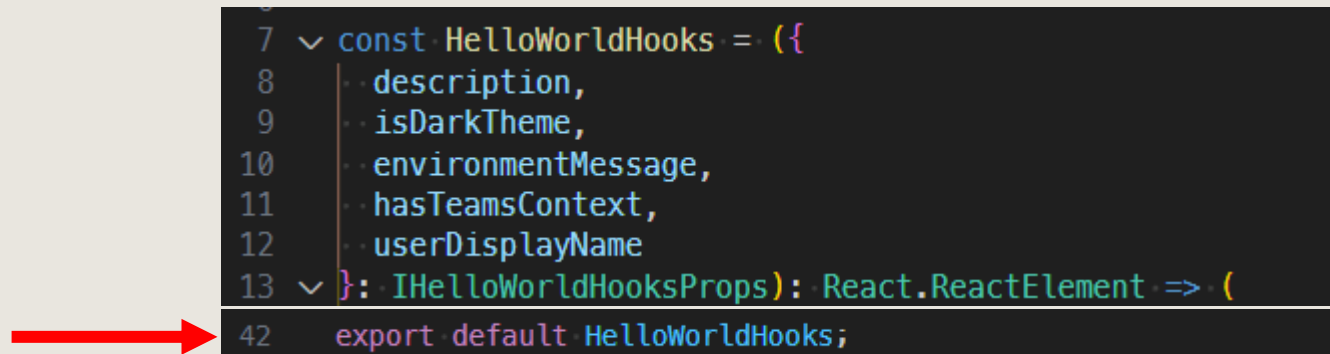
```
12 const LifecycleHooks: React.FC<ILifecycleHooksProps> = (props) => {  
13   const [events, setEvents] = React.useState<string[]>(['[lifecycleHooks] co  
14   const [count, setCount] = React.useState<number>(0);  
15   const [counting, setCounting] = React.useState<boolean>(false);  
16   const timerRef = React.useRef<number | undefined>(undefined);  
17   const didMountRef = React.useRef(false);  
18  
19   > // #region other functions ...  
44  
45   // Simulate componentDidMount/componentWillUnmount  
46   > React.useEffect(() => { ...  
57   }, []);
```


CONVERSION TO FUNCTION COMPONENTS

- Best done right after new project creation
- No changes in the base webpart
- 2 lines + cleanup
 - declaration
 - remove render
 - add export statement



```
export default class HelloWorldClassic extends React.Component<IHelloWorldClassicProps> {  
  public render(): React.ReactElement<IHelloWorldClassicProps> {  
    8    const {  
    9      description,  
    10     isDarkTheme,  
    11     environmentMessage,  
    12     hasTeamsContext,  
    13     userDisplayName  
    14   } = this.props;
```



```
7  const HelloWorldHooks = ({  
8    description,  
9    isDarkTheme,  
10   environmentMessage,  
11   hasTeamsContext,  
12   userDisplayName  
13  }): IHelloWorldHooksProps): React.ReactElement => {  
42  export default HelloWorldHooks;
```

DEMO TIME!

Just warming up!



GAMECHANGERS: useState() & useEffect()

useState()

- replaces this.state, granular state
- no lifecycle methods (useEffects)
- useState declared as a 2 item array
 - first item is variable to hold value
 - second item is function to change the variable value
 - initialization is possible in declaration

```
const [events, setEvents] = React.useState<string[]>(['lifecycle']);
const [count, setCount] = React.useState<number>(0);
const [counting, setCounting] = React.useState<boolean>(false);
```

```
    setCount(0);
    setCounting(false);
```

```
    if (count === 10) {
```

```
        {counting && <div>Counting...</div>}
    }
```

useEffect()

- function that is called after component renders
- uses 2 parameters
 - 1st parameter is callback function that runs when component renders or updates
 - 2nd parameter is dependency array, which determines when useEffect runs
 - no parameter: always runs
 - []: only runs on first render
 - [value]: runs when value changes
- return function can be used to clean up

```
React.useEffect(() => {
  console.log("Effect runs on every render!");
}); // No dependency array means it runs every render
```

```
React.useEffect(() => {
  console.log("Component mounted!");
}, []); // Empty array means it runs only once
```

```
React.useEffect(() => {
  console.log(`Count changed to ${count}`);
}, [count]); // Runs only when `count` updates
```


DEMO TIME!

Still warming up!



useCallback

Preventing unnecessary re-renders in child components.

Particularly useful in cases where you pass functions as props to child components or use them inside event handlers, preventing unnecessary re-renders and improving performance.

Takes two arguments:

- **A function** – The callback you want to memoize.
- **A dependency array** – When dependencies change, the function will be re-created.

```
function Button({ onClick }) {  
  console.log("Button rendered");  
  return <button onClick={onClick}>Click me</button>;  
}  
  
function ParentComponent() {  
  const [count, setCount] = useState(0);  
  
  const increment = useCallback(() => {  
    setCount(prevCount => prevCount + 1);  
  }, []); // Memoized function  
  
  return (  
    <div>  
      <p>Count: {count}</p>  
      <Button onClick={increment} />  
    </div>  
  );  
}
```

useRef

creates a mutable reference to store values without causing re-renders.

It's commonly used for:

- Accessing & modifying DOM elements (e.g., focusing an input field).
- Persisting values across renders (e.g., tracking previous state or storing intervals).
- Avoiding unnecessary state updates (since useRef doesn't trigger re-renders).

```
function InputFocus() {
  const inputRef = useRef<HTMLInputElement>();

  return (
    <div>
      <input ref={inputRef} type="text" />
      <button onClick={() => inputRef.current?.focus()} />
    </div>
  );
}
```

```
return (
  <div style={styles.container}>
    <p>Some content</p>
    <p>More content</p>
    <div ref={inputRef}>
      <input type="text" />
    </div>
  </div>
);
```

```
const [count, setCount] = React.useState<number>(0);
const [counting, setCounting] = React.useState<boolean>(false);
const timerRef = React.useRef<number | undefined>(undefined);
const didMountRef = React.useRef(false);

const startTimer = React.useCallback(() => {
  if (!timerRef.current) {
    timerRef.current = window.setInterval(() => {
      setCount(prev => (prev < 10 ? prev + 1 : prev));
    }, 500);
  }
}, []);

const stopTimer = React.useCallback(() => {
  if (timerRef.current) {
    clearInterval(timerRef.current);
    timerRef.current = undefined;
    setCounting(false);
  }
}, []);

React.useEffect(() => {
  didMountRef.current = true;
  return () => {
    if (timerRef.current) {
      clearInterval(timerRef.current);
      timerRef.current = undefined;
    }
  };
}, []);

React.useEffect(() => {
  if (didMountRef.current) {
    if (count === 10 && counting) {
      stopTimer();
    }
  }
}, [count]);
```


useMemo()

Optimizing performance by memorizing expensive calculations, preventing them from being re-calculated on every render

Takes two arguments:

1. A function – Returns the computed value.
2. A dependency array – The value is only recomputed when dependencies change.

```
function Example() {
  const [count, setCount] = useState(0);

  function SortedList({ items }) {
    const sortedItems = useMemo(() => {
      console.log("Sorting...");
      return [...items].sort();
    }, [items]);

    return <ul>{sortedItems.map((item, index) => <li key={index}>{item}</li>)}</ul>;
  }

  function App() {
    const [items, setItems] = useState(["banana", "apple", "cherry"]);

    function ComputeFactorial({ number }) {
      const factorial = useMemo(() => {
        console.log("Computing factorial...");
        let result = 1;
        for (let i = 2; i <= number; i++) {
          result *= i;
        }
        return result;
      }, [number]);

      return <p>Factorial of {number}: {factorial}</p>;
    }

    return (
      <div>
        <input type="text" value={query} />
        <button onClick={query} />
        <ul>{filteredItems.map(item => <li key={item}>{item}</li>)}</ul>
      </div>
    );
  }
}
```

useContext()

Allows components to access global state without having to pass props manually at every level.

It's useful for managing theme settings, authentication status, user preferences, or any shared state across multiple components.

How It Works

1. Create a Context – Define shared state using `createContext()`.
2. Provide a Context Value – Use `Context.Provider` to pass data down.
3. Consume Context in Components – Use `useContext()` to access the shared state.

```
import { createContext, useContext } from "react";

const ThemeContext = createContext("light"); // Default value: "light"

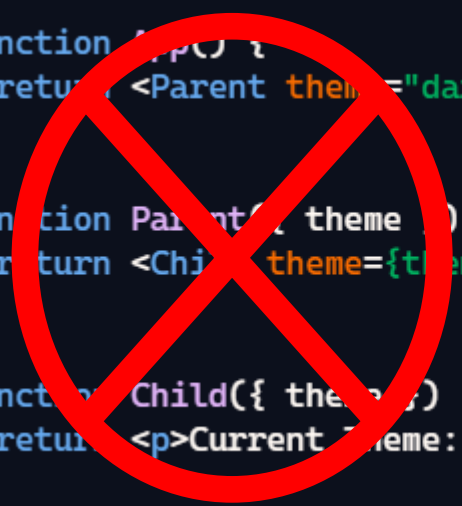
function ChildComponent() {
  const theme = useContext(ThemeContext); // Access context value
  return <p>Current Theme: {theme}</p>;
}

function ParentComponent() {
  return (
    <ThemeContext.Provider value="dark">
      <ChildComponent />
    </ThemeContext.Provider>
  );
}

function App() {
  return (
    <ParentComponent theme="dark" />
  );
}

function Child({ theme }) {
  return <p>Current Theme: {theme}</p>;
}

function App() {
  return (
    <AuthContext.Provider value={{ name: "Alice" }}>
      <Profile />
    </AuthContext.Provider>
  );
}
```



Custom Hooks

Encapsulates reusable logic into functions, making code cleaner and more modular.

They're just JavaScript functions that use built-in hooks like `useState()`, `useEffect()`, or `useContext()` to provide specific functionality.

Why?

- Avoid repeated logic across multiple components
- Improve code readability and organization
- Encapsulate side effects cleanly

```
import { useState, useEffect } from "react";

function useWindowSize() {
  const [size, setSize] = useState({
    width: window.innerWidth,
    height: window.innerHeight,
  });

  useEffect(() => {
    const handleResize = () => {
      setSize({ width: window.innerWidth, height: window.innerHeight });
    };

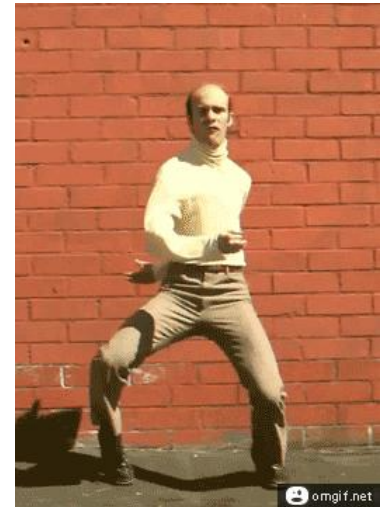
    window.addEventListener("resize", handleResize);
    return () => window.removeEventListener("resize", handleResize);
  }, []);

  return size;
}

function Component() {
  const { width, height } = useWindowSize();
  return <p>Window size: {width}x{height}</p>;
}
```


DEMO TIME!

PUT ON YOUR DANCING SHOES AND LET'S HAVE SOME FUN!



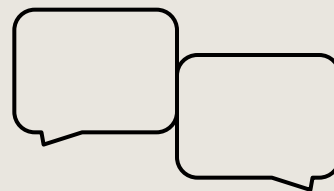
Recap: Hooks at a Glance

What each hook is for — and where it earns its place

useState	Local, reactive state that re-renders on change. Forms, toggles, counters, loading/error flags.
useEffect	Side effects after render, with cleanup. Data fetching, subscriptions, event listeners, timers.
useContext	Shared data without prop drilling. Theme, authenticated user, locale, feature flags.
useRef	Mutable value or DOM access without re-rendering. Focus, measuring, timer IDs, previous values.
useMemo	Caches expensive results and keeps referential stability. Filtering/sorting large lists, heavy transforms.
useCallback	Memoizes a function's identity. Handlers for React.memo children; stable hook dependencies.
Custom Hooks	Compose hooks to reuse stateful logic. useFetch, useDebounce, useLocalStorage, useForm.

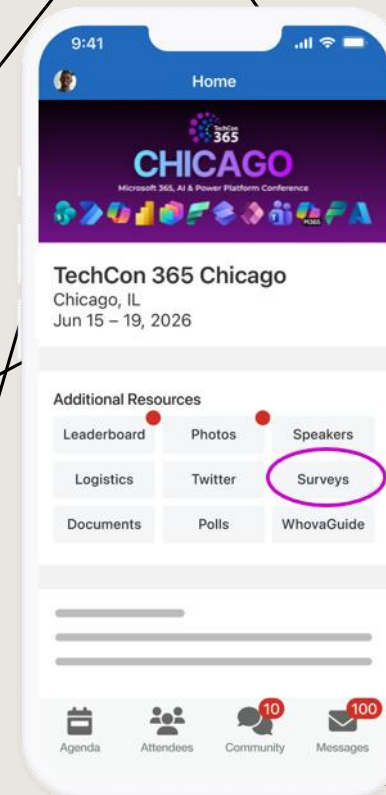
References

- **This presentation** — <https://github.com/DonKirkham/Presentations/tree/main/2026/2026-06-TechCon-Chicago>
- **Demo project** — <https://github.com/donkirkham/hooksdemo>
- **React Function Components** — <https://react.dev/reference/react/Component#alternatives>
- **React Hooks** — <https://react.dev/reference/react/hooks>



Q&A

Questions?



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